Basis for infinite dimension vector space

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1 Brief Introduction

When student study linear algebra in year 1. For math majored student, they might started from vector space, linear independent , basis and so on. From Maximal Principle(which equivlemt to Zorn's lemma and Axiom of Choice), we can directly prove that every vector space has a basis. Therefore, that may lead students to think more about it. Such as the basis of all continuous function, all differentiable function and so on. However, those students will quickly find out that it is quite a impossible tesk.
I find every text book of linear algebra in my university, and I am sadly to tell that there are no such solution in those books. Such that i turned my sight into Functional Analysis and I finally found those solutions.
Here, the reason I wrote this blog is for students like me who determined in the curios of the completment of Mathematics and eager to find out the truth.

2 Hamel-Basis

The basis we used in linear algebra is actually **Hamel-Basis**. Here I will first prove the concepts that every vector space has a basis(Hamel-basis). Since most students have studied most of the basic concepts, so I will skip the part such as the definition and the basic properties.
To prove the corollary, we need some definiitons and theorems.

Definition 2.1 (Maximal Member)**.** *Let $\mathbb{F}$ be a family of sets. A member $M \in \mathbb{F}$ is called maximal if M is contained in no other member of $\mathbb{F}$*

Definition 2.2 (Chain)**.** *A family of sets $\mathcal{C}$ is called a chain if for each pair of sets A,B in $\mathcal{C}$, we have either $A \subseteq B$ or $B \subseteq A$.*

Lemma 2.3 (Maixmal Principle)**.** *Let $\mathbb{F}$ be a family of stes, for every $\mathcal{C} \subseteq \mathbb{F}$. $\exists\, \mathbb{F} \in \mathbb{F}$ such that $\mathbb{F}$ contains every member in $\mathcal{C}$, then $\mathbb{F}$ has a max member.*

Definition 2.4. *Let S be a subet of a vector space V. A maximal linearly independent subset of S is a subset B of S satisfying both of the following conditions.*
(a) B is linearly independent
(b) The only linearly independent subset of S that contains B is B itself.

Theorem 2.5. *Let Vbe a vector space and S a subset that generates V. If β is a maximal linearly independent subset of S, then β is a basis for V.*

Theorem 2.6. *Let S be a linearly independent subset of a vector space V. There exists a maximal linearly independent subset of V that contains S.*

Proof. Let $\mathbb{F}$ denote the family of all linearly independent subsets of V that contain S. In order to show that $\mathbb{F}$ contains a maximal element, we must show that if $\mathcal{C}$ is a chain in $\mathbb{F}$, then there exists a member U of $\mathbb{F}$ that contains each member of $\mathcal{C}$. We claim that U , the union of the members of $\mathcal{C}$, is the desired set. Clearly U contains each member of $\mathcal{C}$, and so it suffices to prove that U $\in$ F (i.e., that U is a linearly independent subset of V that contains S). Because each member of $\mathcal{C}$ is a subset of V containing S, we have S $\subseteq$ U $\subseteq$ V. Thus we need only prove that U is linearly independent. Let u_1, u_2, . . . , u_n be in U and a_1, a_2, . . . , a_n be scalars such that $a_1u_1 + a_2u_2 + \cdots + a_nu_n = 0$. Because $u_i \in$ U for i = 1, 2, . . . , n, there exists a set A_i in $\mathcal{C}$ such that $u_i \in A_i$. But since $\mathcal{C}$ is a chain, one of these sets, say A_k, contains all the others. Thus $u_i \in A_k$ for i = 1, 2, . . . , n. However, A_k is a linearly independent set; so $a_1u_1 + a_2u_2 + \cdots + a_nu_n = 0$ implies that $a_1 = a_2 = \cdots = a_n = 0$. It follows that U is linearly independent.
The maximal principle implies that $\mathbb{F}$ has a maximal element. This element is easily seen to be a maximal linearly independent subset of V that contains S. □

Theorem 2.7. *Every vector space has a basis*

Proof. Let $\mathbb{F}$ be the family of all linear independent subsets of V that contains S
Claim:$\mathbb{F}$ has a maximal elements.Such that suffice to show: For every $\mathcal{C} \subseteq \mathbb{F}$, $\exists$ F $\in$ s.t. every member of $\mathcal{C}$ is contained in F. To prove this, set F=the union of all members in $\mathcal{C}$
Need to show: $\mathbb{F}$ is linearly independent.
Let u_1,u_2,...$u_k \in \mathbb{F}$, such that $\exists$ A_1,...,$A_K \in \mathcal{C}$. Such that u_1,...,$u_k \in A_k$. Since $\mathcal{C}$ is a chain, we conclude that$\exists$ $A_i \subseteq A_j$

According to replacement theorem, every basis for an infinite-dimensional vector space has the same cardinality. □

3 The changing of the view

From the previous section, we used zorn's lemma(Axiom of choice) to prove that every vector space has a basis. For example, for $\mathbb{R}^n$, we can easily find basis equals to $e_1, e_2, ..., e_n$. However, what about $\mathbb{C}_[0, 1]$?

4 Schaedler-Basis

4.1 Metric space

Definition 4.1 (norm)**.** *A norm on a (real or complex) vector space X is a real-valued function on X whose value at an x $\in$X is denoted by* $\|x\|$

1. $\|x\| \geq 0$
2. $\|x\| = 0 \iff x = 0$
3. $\|\alpha x\| = |\alpha| \, \|x\|$ *for all* $\alpha \in \mathbb{K}, x \in X$
4. $\|x + y\| \leq \|x\| + \|y\|$

*The pair $(X, \|\cdot\|)$ is called a **normed space**.*

Definition 4.2 (**Cauchy sequence,completeness**). *A sequence (x_n) in a metic space X=(X,d) is said to be Cauchy (or fundamental) if for every $\epsilon > 0$ there is an $N = N(\epsilon)$ such that*

$$d(x_m, x_n) < \epsilon$$

for every m,n>N
*The space X is said to be **complete** if every Cauchy sequence in X converges(that is, has a limit which is an element of X).*
Expressed in terms of completeness, the Cauchy convergence criterion implies the following.

Theorem 4.3 (Complete subspace). *A subspace M of a complete metric space X is itself complete if and only if the set M is closed in X.*

Proof. Let M be complete. For every $x \in \bar{M}$ there is a sequence (x_n) in M which converges to x. Since (x_n) is Cauchy and M is complete, (x_n) converges in M, the limit being unique . Hence $x \in M$. This proves that M is closed because $x \in \bar{M}$ was arbitrary as well.

Conversely, let M be closed and (x_n) Cauchy in M. Then $x_n \longrightarrow x \in X$, which implies $x \in \bar{M}$, and $x \in M$ since $M = \bar{M}$ by assumption. Hence the arbitrary Cauchy sequence (x_n) converges in M, which proves completeness of M. □

This theorem is very useful, and we shall need it quite often. Example of the space C consists of all convergent sequences of complex numbers includes the first application, which is typical.

Definition 4.4 (Banach space). *The banach space is a **complete** normed space.*

After we give these definitions, we find out that despite the vector space we have learnt in the linear algebra course, we actually put the concepts including **length , limit**(which refers to completeness) into a vector space. This is like we first build a rough algebraic structure **Vector Space** which is grand stucture that we can nearly put everything in it. However vector space is too big and too rough, such that it almost have nothing to do with some kind of structures which need more stronger properties. Now, in order to analyse a better structure, we will put another important thing in it ,which is called **Inner Product**.This will lead us to a nearly "perfect" space **Hlbert Space** $\mathscr{H}$. Why I put it in this blog have several reasons. The main reason is from our previous study, the basis we have met are almost all orthogonal(like x,y-axis), for C[a,b], unfortunately, wecan't use it.

Definition 4.5 (Inner Product Space). *Let V be a vector space over a field $\mathbb{F}$ (typically $\mathbb{R}$ or $\mathbb{C}$). An **inner product** on V is a map $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$ that satisfies the following properties for all $x, y, z \in V$ and scalar $\alpha \in \mathbb{F}$:*

1. ***Conjugate Symmetry:***

$$\langle x, y \rangle = \overline{\langle y, x \rangle}$$

   (Note: In a real vector space, this implies symmetry: $\langle x, y \rangle = \langle y, x \rangle$.)

2. ***Linearity in the first argument:***

$$\langle \alpha x + y, z \rangle = \alpha \langle x, z \rangle + \langle y, z \rangle$$

3. ***Positive Definiteness:***

$$\langle x, x\rangle \geq 0, \quad \textit{and} \quad \langle x, x\rangle = 0 \iff x = 0$$

A vector space equipped with an inner product is called an Inner Product Space (or Pre-Hilbert Space).

Definition 4.6 (Hilbert Space)**.** *A* ***Hilbert Space*** $\mathscr{H}$ *is an inner product space that is* ***complete*** *with respect to the norm induced by its inner product.*

The induced norm is defined as:

$$\|x\| = \sqrt{\langle x, x\rangle}$$

"Complete" *means that every Cauchy sequence in* $\mathscr{H}$ *converges to a limit that is also within* $\mathscr{H}$*.*

So in Functional Analysis, we always have a example $\mathbb{C}$[a,b] which is the set of all continuous function. Now we examine the **Completeness** of $\mathbb{C}$

The space $\mathbb{C}$[a,b] with the metric

Definition 4.7 (Function space $\mathbb{C}[a, b]$)**.** *As a set X we take the real-valued functions x,y,... which are functions of an independent real variables t and are defined and continuous on a given closed interval J=[a,b]. Choosing the metric defined by*

d(x,y)= max